

Synthesis, Portfolio Writing, and Final Presentations

Make database skill visible through safe demonstrations, reproducible artifacts, and honest technical stories.

15

The course built verbs, not a list of product names

**Model. Query. Constrain. Authorize. Measure. Recover.
Explain.**

BUILD

Represent facts, relationships, rules, and workloads in relational and document systems.

OPERATE

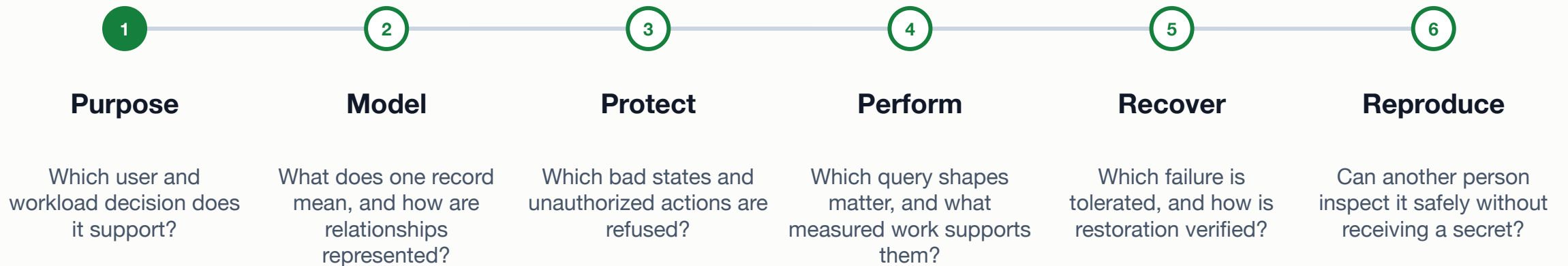
Observe identities, plans, locks, failures, backups, restores, and cloud boundaries.

COMMUNICATE

Connect every claim to reproducible results, a tradeoff, and an honest limit.

Review any data system through six connected questions

A system is credible when meaning, behavior, protection, and recovery form one chain.



A tool claim becomes credible when scope and results are specific

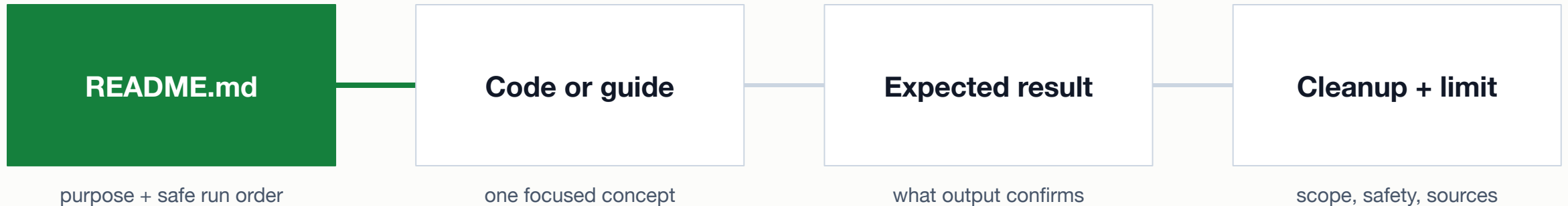
Tool inventory

- Used PostgreSQL, MongoDB, Atlas, Supabase, and Python
- Added indexes and security
- Made a backup

Specific skill statement

- Built named constraints and verified expected failures
- Compared unchanged query plans before and after a composite index
- Restored a logical artifact elsewhere and checked structure, data, and behavior

A useful GitHub project teaches one technical sequence



A focused operations artifact can demonstrate more skill than a large app whose data decisions are hidden.

Publication safety is part of technical quality

- 1 Search every file and notebook output for credentials, tokens, private hosts, and URLs containing secrets.

- 2 Remove real personal or regulated data and use synthetic or appropriately licensed data instead.

- 3 Name disposable targets, label destructive operations, and test the documented run order in a clean path.

- 4 Add source and license attribution, then verify the review URL without relying on your signed-in session.

Final lab: Teach one database concept in GitHub

Create one small, safe artifact that another beginner can understand and reproduce.

- Choose one course concept and state the problem, prerequisite, and observable learning objective.
- Publish a README plus one code artifact or a detailed Markdown guide with concrete code examples.
- Run the publication-safety check, verify access, and explain what one output demonstrates and does not confirm.

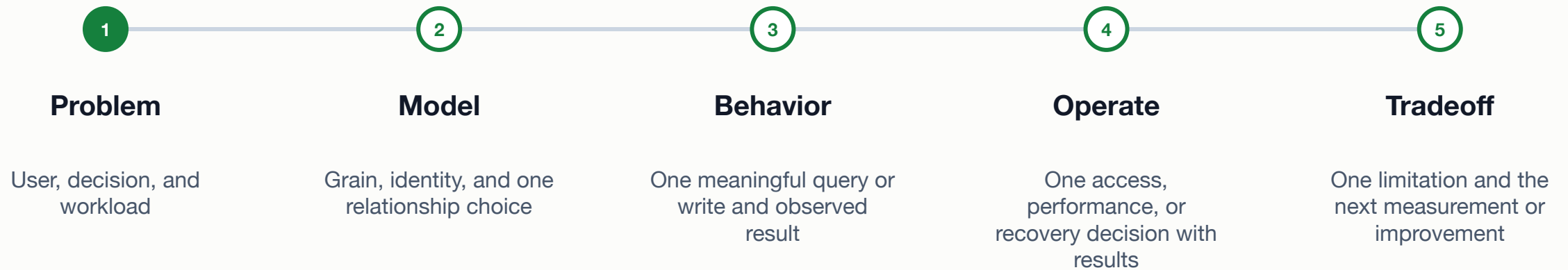
SUBMIT ONE THING / one public or instructor-accessible artifact URL in Brightspace

Present the project as a decision story

Show why the system exists, what you chose, which results supports it, and where its limits begin.

A focused demonstration follows one technical arc

The audience should be able to repeat the claim in one sentence afterward.



STAR-R turns a lab into an honest incident story

Situation → Task → Action → Result → Reflection

SPECIFIC

Name the controlled context, your responsibility, and the results you personally inspected.

OBSERVABLE

Describe the result as a changed state, output, plan, denial, or verified restore.

REFLECTIVE

State the limitation and what production results or control would come next.

Career language should match the work actually performed

Interview question	Weak answer	Specific technical basis	Honest limit
Why this index?	Indexes make it faster	same query, changed plan and examined work	fixture is not production load
PostgreSQL or MongoDB?	MongoDB is flexible	workload, relationships, integrity, access pattern	choice could change with measurements
How did you secure it?	I used cloud security	least privilege plus allowed and denied tests	tested identities and paths only
Can you recover it?	There is a backup	separate restore plus structure/data/behavior checks	RPO, RTO, and full config remain

Honest scope is stronger than inflated certainty

Overclaim

- Production-ready, secure, scalable, and fault tolerant
- Expert in database administration
- Guaranteed exactly-once processing

Credible scope

- Reproduced and verified the mechanism in an isolated course environment
- Observed result supports the tested query, identity, failure, and target
- Production next step is named where scale or operations remain untested

Explain one technical decision as a complete story



`"I reproduced ___ in an isolated lab, observed ___, and would next test ___."`

Specific scope is stronger than inflated certainty.

Your strongest course outcome is a repeatable way to reason

State the workload, choose a mechanism, observe results, name the tradeoff, and preserve the limit.

- What did you personally build or investigate?

- Which observation shows it worked or failed as expected?

- What would you inspect or control next in production?